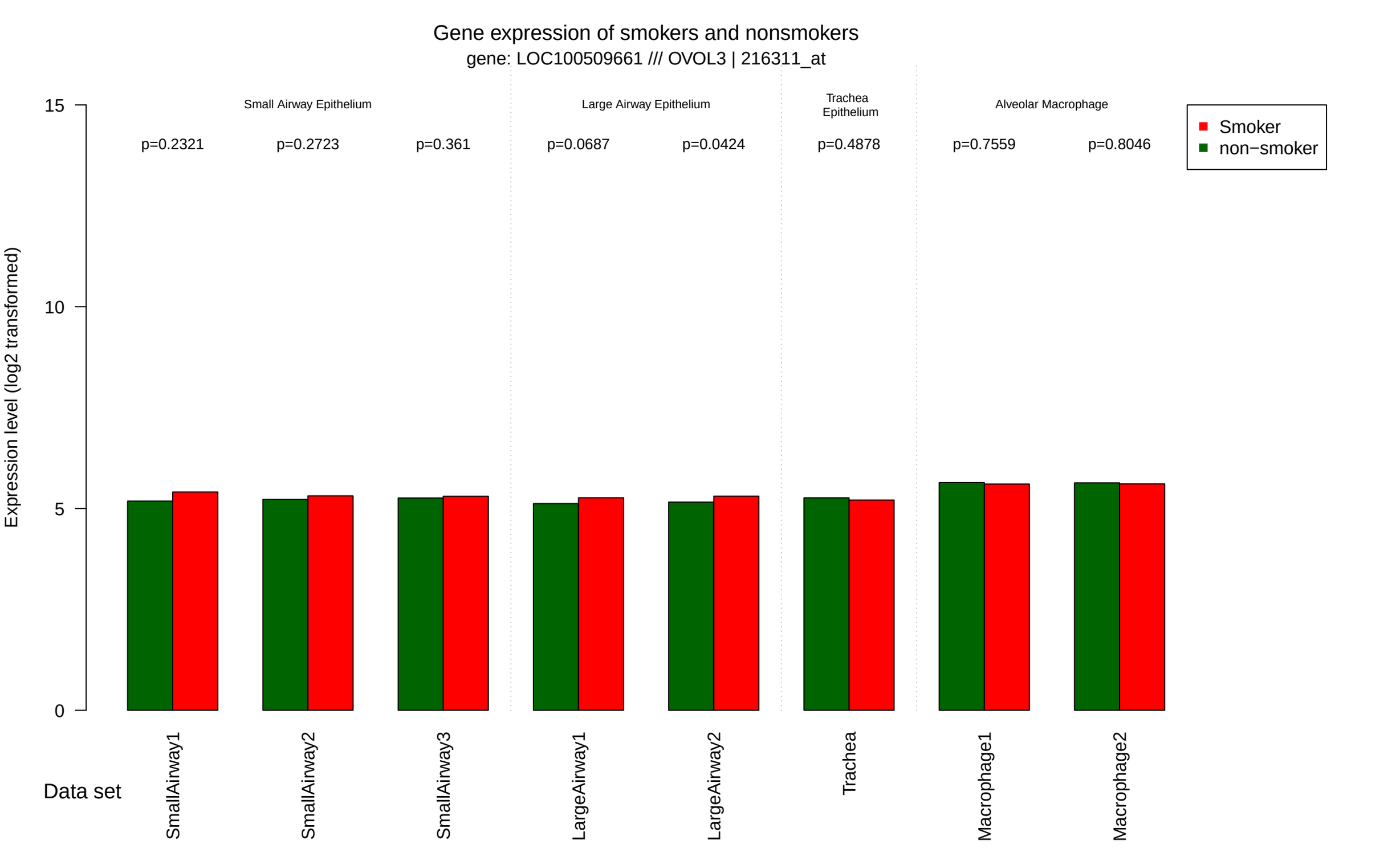
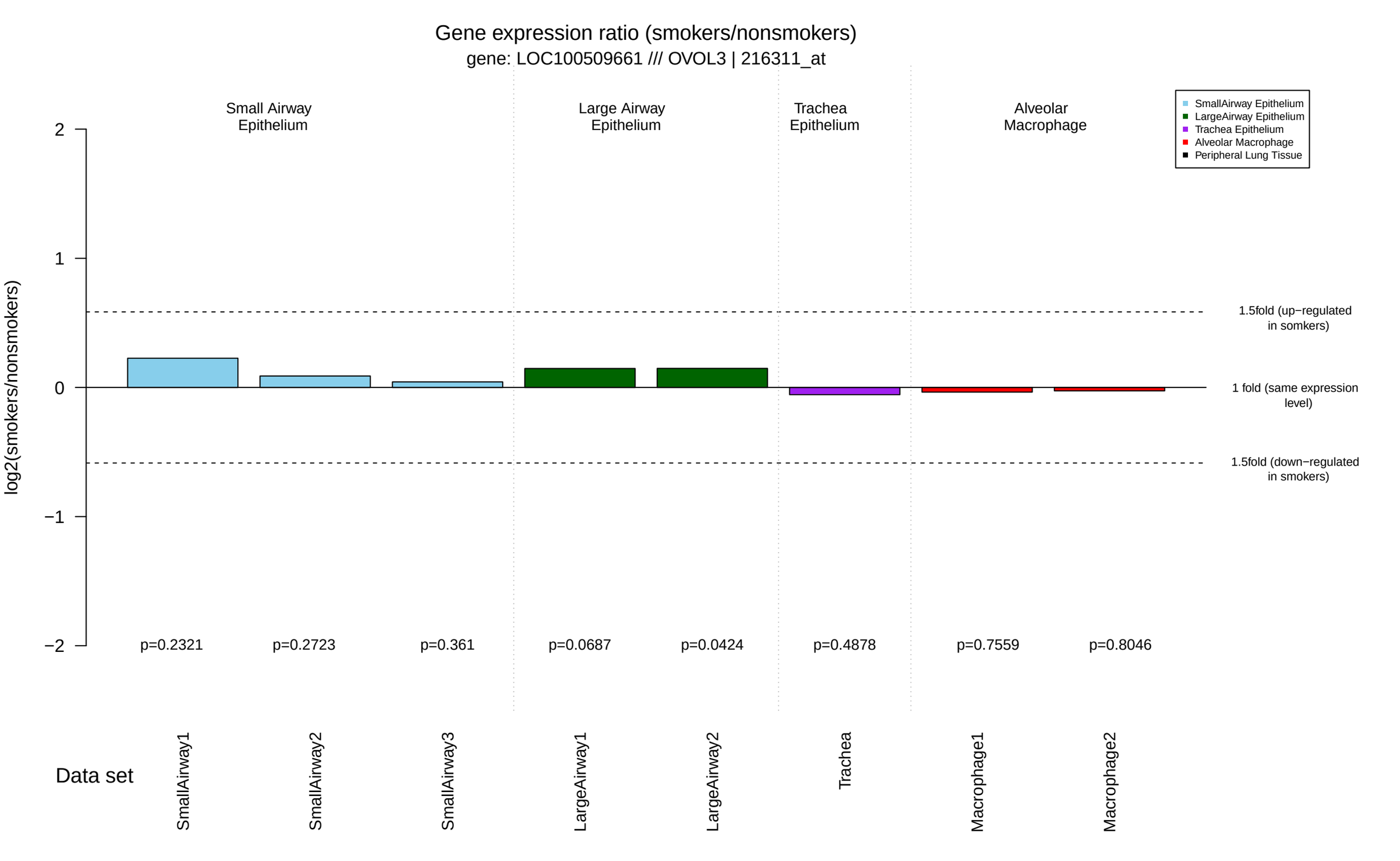
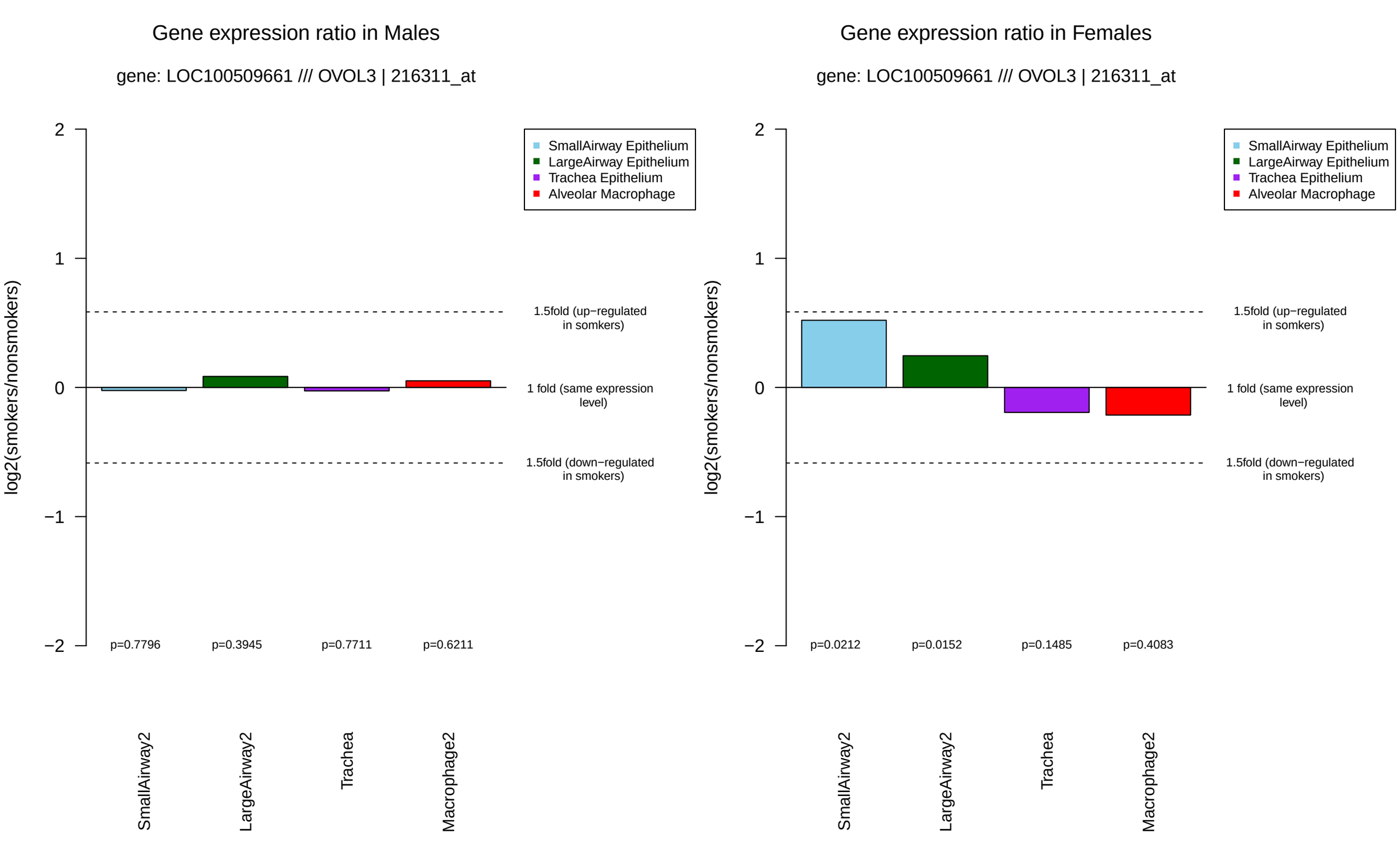
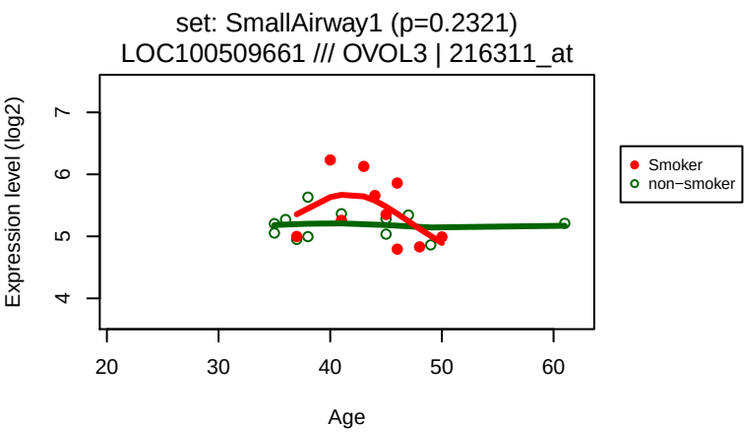
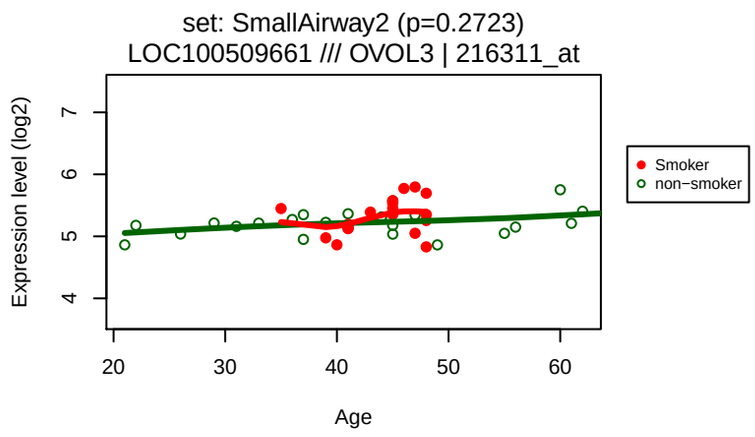
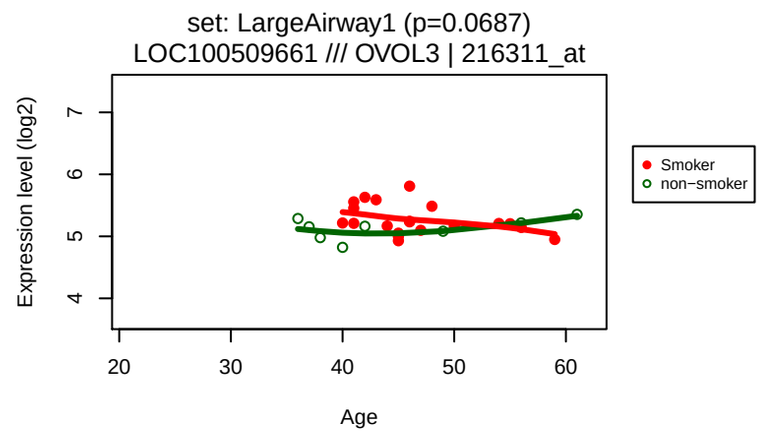

Gene expression of smokers and nonsmokers
gene: LOC100509661 /// OVOL3 | 216311_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2321
p=0.2723
p=0.361
p=0.0687
p=0.0424
p=0.4878
p=0.7559
p=0.8046
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

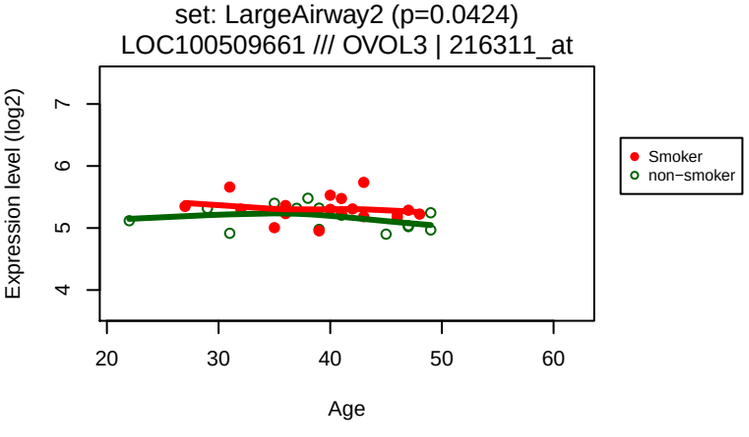

Gene expression ratio (smokers/nonsmokers)
gene: LOC100509661 /// OVOL3 | 216311_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2321
p=0.2723
p=0.361
p=0.0687
p=0.0424
p=0.4878
p=0.7559
p=0.8046
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

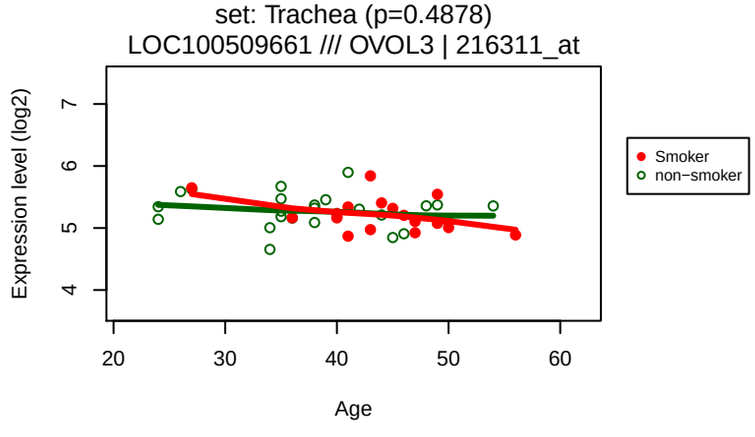

Gene expression ratio in Males
gene: LOC100509661 /// OVOL3 | 216311_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7796
p=0.3945
p=0.7711
p=0.6211
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC100509661 /// OVOL3 | 216311_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0212
p=0.0152
p=0.1485
p=0.4083
SmallAirway2
LargeAirway2
Trachea
Macrophage2

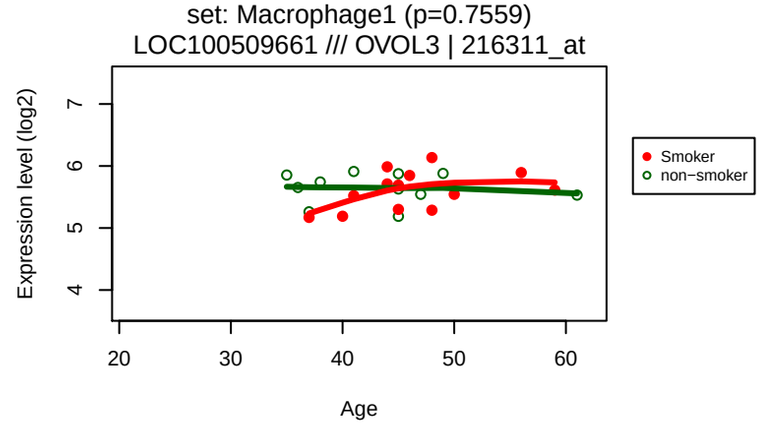

set: SmallAirway1 (p=0.2321)
LOC100509661 /// OVOL3 | 216311_at
Expression level (log2)
Age
Smoker
non−smoker

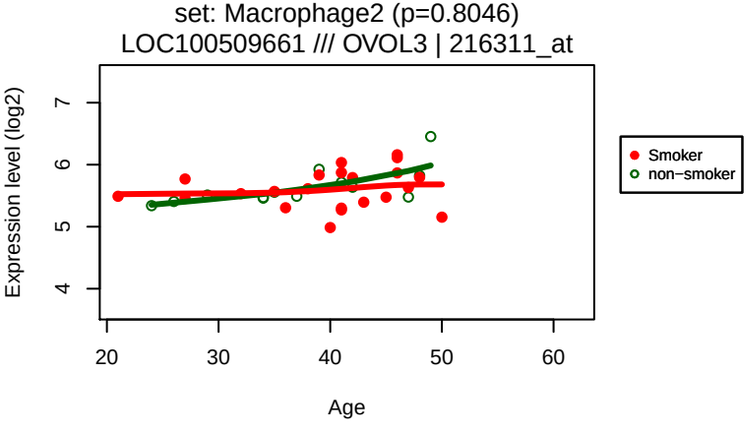

set: SmallAirway2 (p=0.2723)
LOC100509661 /// OVOL3 | 216311_at
Expression level (log2)
Age
Smoker
non−smoker

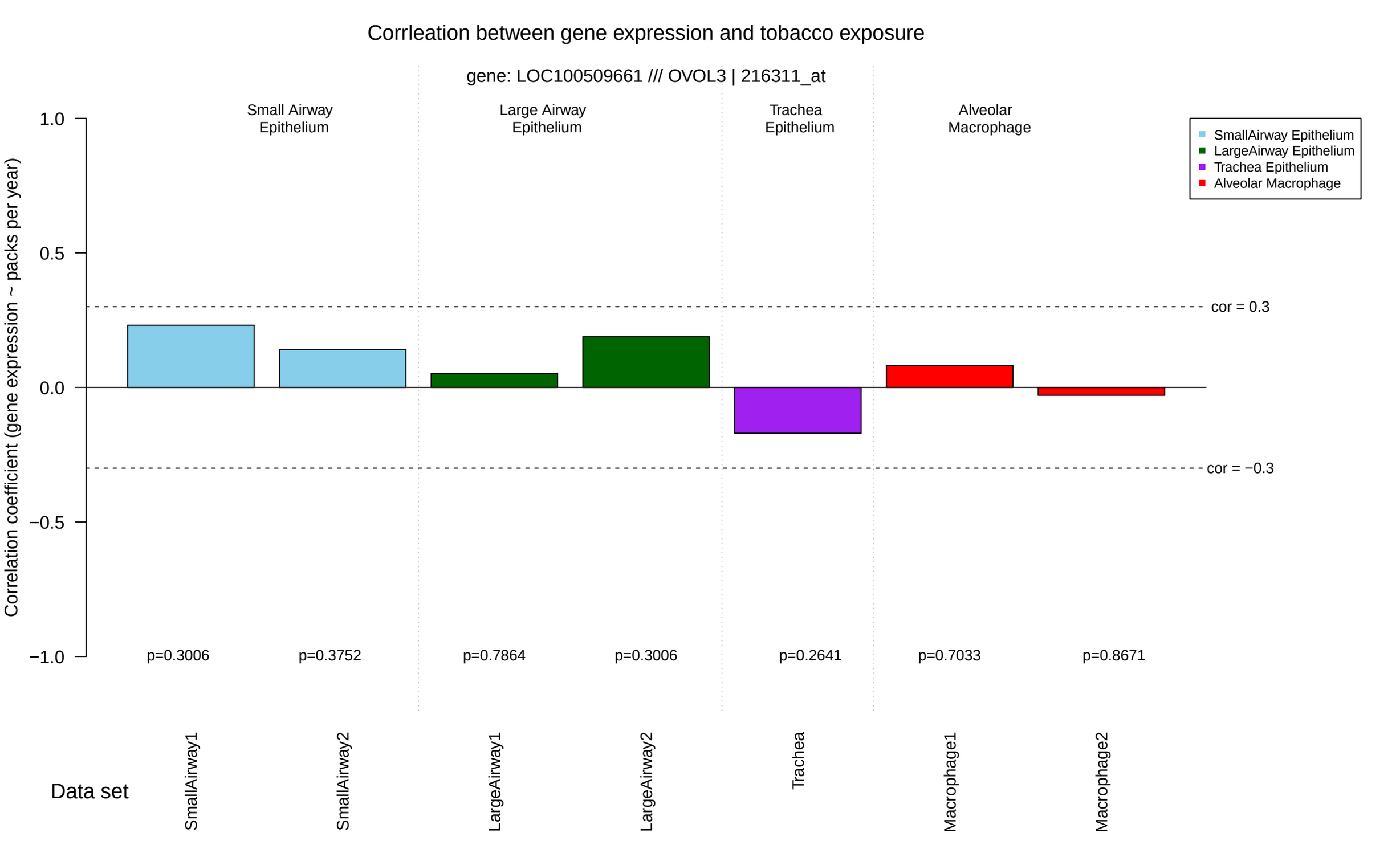

set: LargeAirway1 (p=0.0687)
LOC100509661 /// OVOL3 | 216311_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0424)
LOC100509661 /// OVOL3 | 216311_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4878)
LOC100509661 /// OVOL3 | 216311_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7559)
LOC100509661 /// OVOL3 | 216311_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8046)
LOC100509661 /// OVOL3 | 216311_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC100509661 /// OVOL3 | 216311_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3006
p=0.3752
p=0.7864
p=0.3006
p=0.2641
p=0.7033
p=0.8671
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2